Q 1: Given an array of strings words, return the first palindromic string in the array. If there is no such string, return an empty string "". A string is palindromic if it reads the same forward and backward.

Example 1:

Input: words = ["abc","car","ada","racecar","cool"]

Output: "ada"

Explanation: The first string that is palindromic is "ada".

Note that "racecar" is also palindromic, but it is not the first.

Example 2:

Input: words = ["notapalindrome","racecar"]

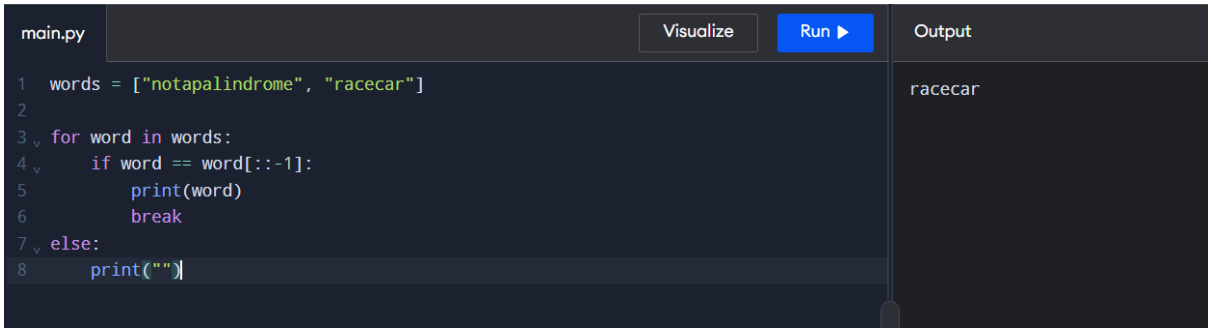
Output: "racecar"

Explanation: The first and only string that is palindromic is "racecar".

Example 1:



Example 2:



Q 2: You are given two integer arrays nums1 and nums2 of sizes n and m, respectively. Calculate the following values: answer1 : the number of indices i such that nums1[i] exists in nums2. answer2 : the number of indices i such that nums2[i] exists in nums1 Return [answer1,answer2].

Example 1:

Input: nums1 = [2,3,2], nums2 = [1,2]

Output: [2,1]

Explanation:

Example 2:

Input: nums1 = [4,3,2,3,1], nums2 = [2,2,5,2,3,6]

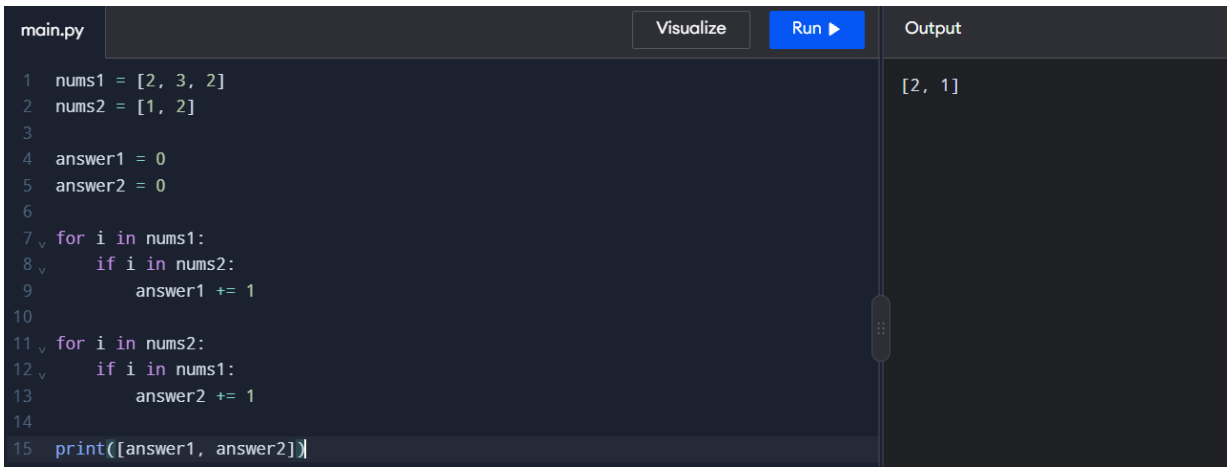
Output: [3,4]

Explanation:

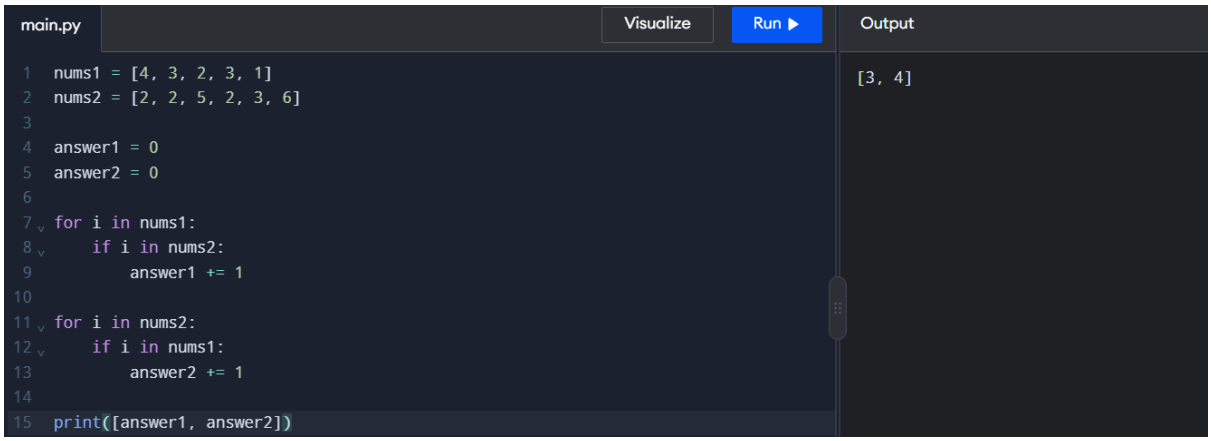
The elements at indices 1, 2, and 3 in nums1 exist in nums2 as well. So answer1 is 3.

The elements at indices 0, 1, 3, and 4 in nums2 exist in nums1. So answer2 is 4.

Example 1:



Example 2:



Q 3: You are given a 0-indexed integer array nums. The distinct count of a subarray of nums is defined as: Let nums[i..j] be a subarray of nums consisting of all the indices from i to j such that 0 <= i <= j < nums.length. Then the number of distinct values in nums[i..j] is called the distinct count of nums[i..j]. Return the sum of the squares of distinct counts of all subarrays of nums. A subarray is a contiguous non-empty sequence of elements within an array.

Example 1:

Input: nums = [1,2,1]

Output: 15

Explanation: Six possible subarrays are:

[1]: 1 distinct value

[2]: 1 distinct value

[1]: 1 distinct value

[1,2]: 2 distinct values

[2,1]: 2 distinct values

[1,2,1]: 2 distinct values

The sum of the squares of the distinct counts in all subarrays is equal to 12 + 12 + 12 + 22 + 22 + 22 = 15.

Example 2:

Input: nums = [1,1]

Output: 3

Explanation: Three possible subarrays are:

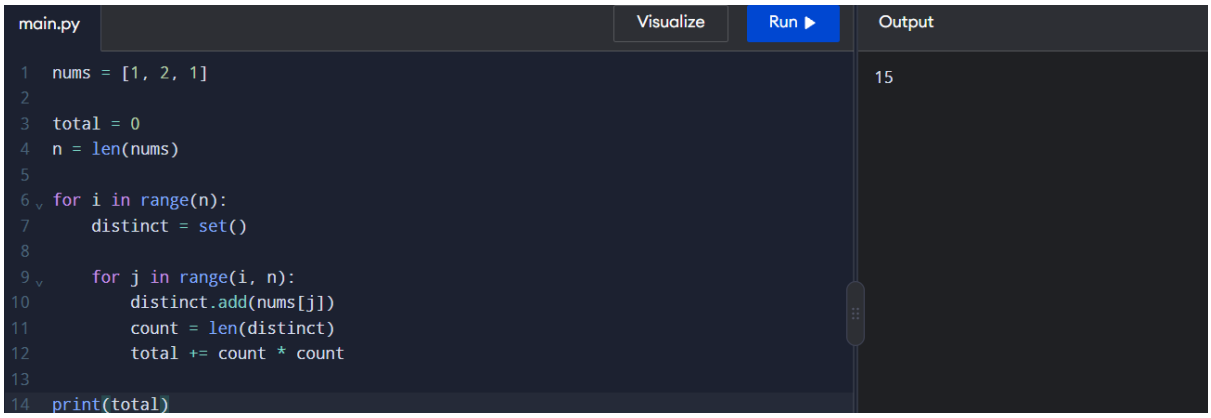
[1]: 1 distinct value

[1]: 1 distinct value

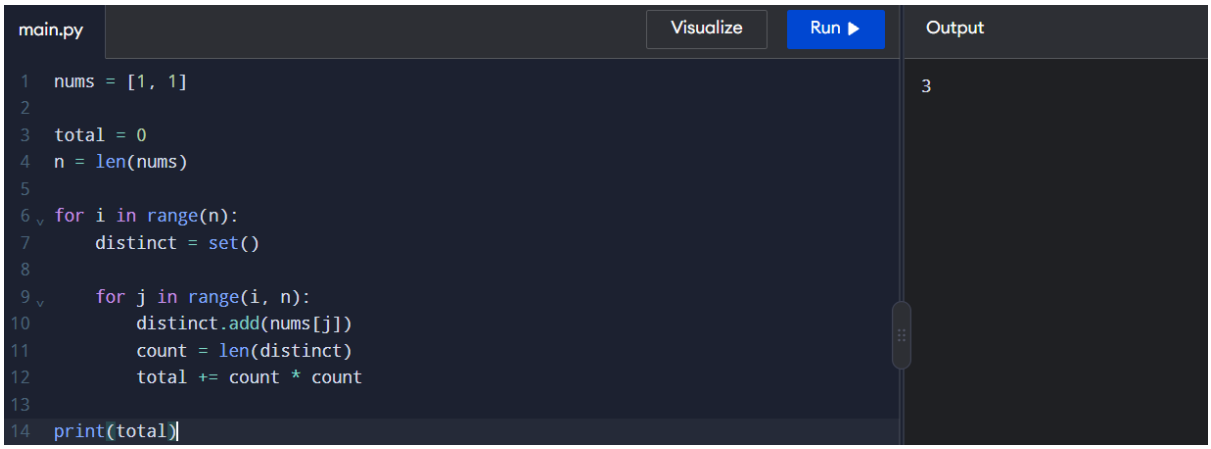
[1,1]: 1 distinct value

The sum of the squares of the distinct counts in all subarrays is equal to 12 + 12 + 12 = 3.

Example 1:



Example 2:



Q 4: Given a 0-indexed integer array nums of length n and an integer k, return *the number of pairs* (i, j) *where* 0 <= i < j < n, *such that* nums[i] == nums[j] *and* (i * j) *is divisible by* k.

Example 1:

Input: nums = [3,1,2,2,2,1,3], k = 2

Output: 4

Explanation:

There are 4 pairs that meet all the requirements:

- nums[0] == nums[6], and 0 * 6 == 0, which is divisible by 2.

- nums[2] == nums[3], and 2 * 3 == 6, which is divisible by 2.

- nums[2] == nums[4], and 2 * 4 == 8, which is divisible by 2.

- nums[3] == nums[4], and 3 * 4 == 12, which is divisible by 2.

Example 2:

Input: nums = [1,2,3,4], k = 1

Output: 0

Explanation: Since no value in nums is repeated, there are no pairs (i,j) that meet all the requirements.

Example 1:

main.py Visualize Run Output

```
nums = [3, 1, 2, 2, 2, 1, 3]
k = 2

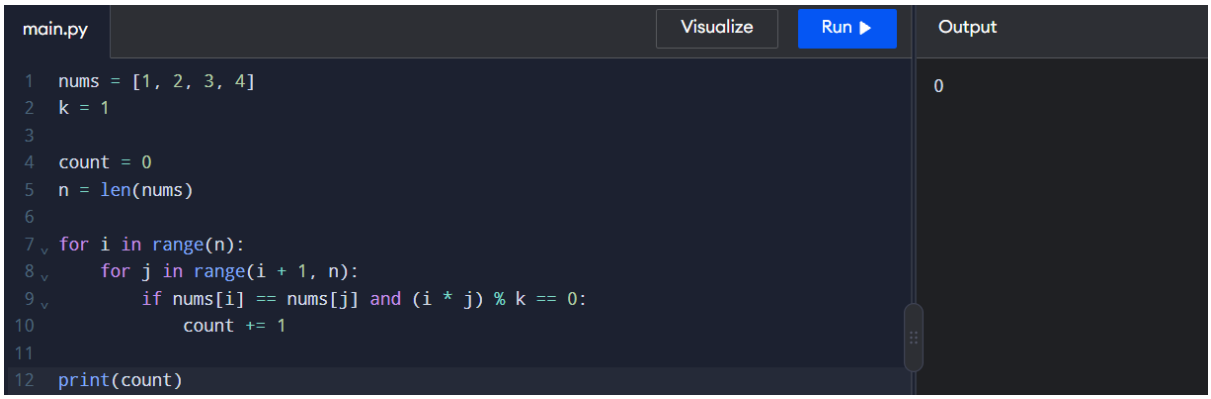
count = 0
n = len(nums)

for i in range(n):
    for j in range(i + 1, n):
        if nums[i] == nums[j] and (i * j) % k == 0:
            count += 1

print(count)
```

```
4
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Example 2:



Q 5. Write a program FOR THE BELOW TEST CASES with least time complexity Test Cases: -

Input: {1, 2, 3, 4, 5} Expected Output: 5

Input: {7, 7, 7, 7, 7} Expected Output: 7

Input: {-10, 2, 3, -4, 5} Expected Output: 5

